Trainings

General Information

Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

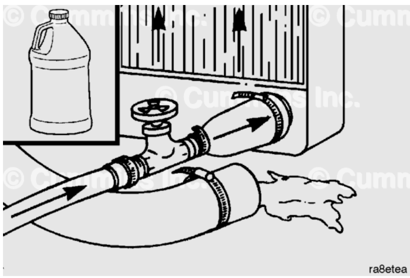
Drain

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.

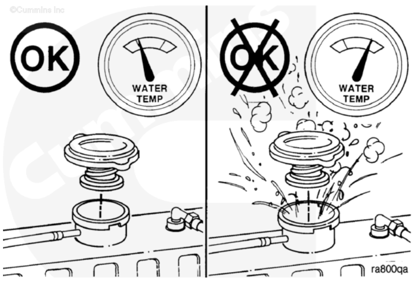
Every 6000 hours or 2 years of operation, whichever comes first, change the coolant and antifreeze.

The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.



⚠ WARNING ⚠

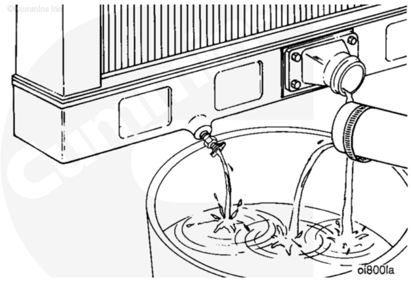
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Do **not** allow the cooling system to dry out.

Drain the cooling system as follows:

- Open the radiator draincock.
- Remove the lower radiator hose.
- Do **not** remove the coolant filter.

Flush

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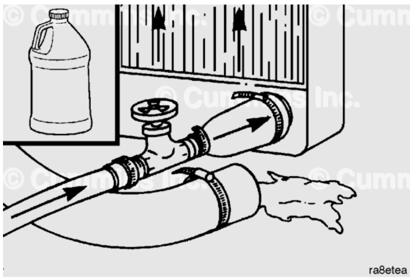
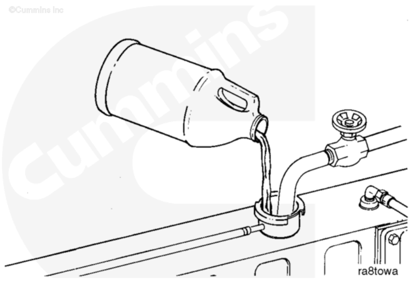


The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.



Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore™ Plus, or the equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

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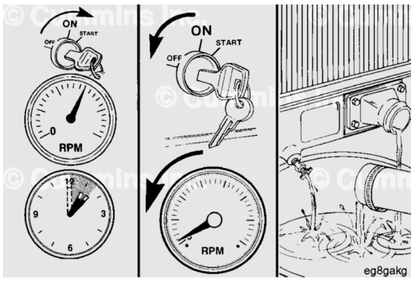
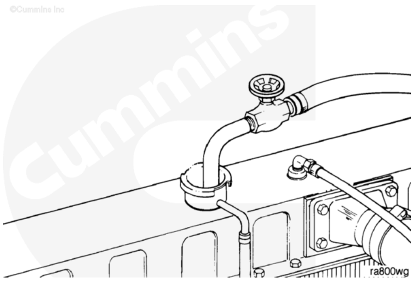


Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Shut the engine OFF, allow to cool to 50° C [120°F], and drain the cooling system.

Fill the cooling system with clean water.



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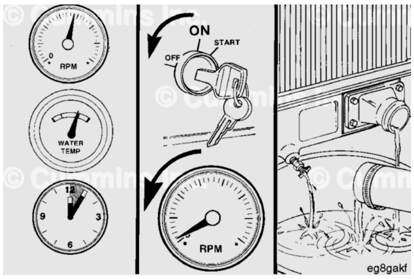


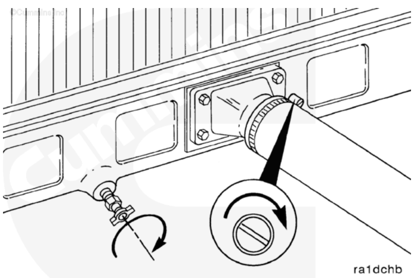
Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

If the water being drained is still dirty, the system **must** be continually flushed until the water is clean.

Fill

Note : This method is to be used when the vehicle/equipment or engine is **not** equipped with a quick disconnect fitting in the cooling system package. If the engine or vehicle/equipment is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method **must** be used.



Close the radiator draincocks.

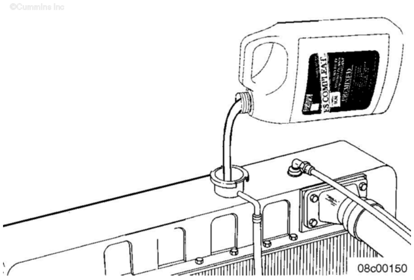
Install the lower radiator hose(s).

Tighten the hose clamps.

Torque Value: 5 n•m [40 in-lb]

⚠ CAUTION ⚠

Do not add coolant to a hot engine. Engine castings can be damaged. Allow the engine to cool below 50°C [120°F] before adding coolant.



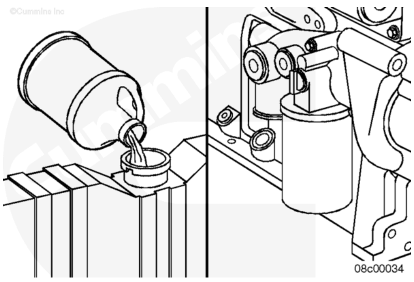
Cummins Inc. recommends using either a 50/50 mixture of good quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant **must** meet TMC RP329 or TMC RP330 specifications.

Good quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as ($CaCO_3$ + $MgCO_3$)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO_4)

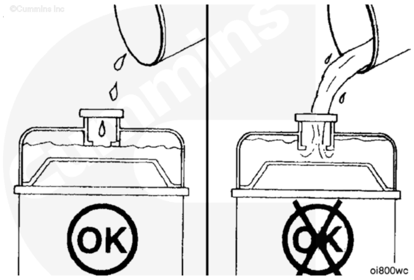
Cummins Inc. recommends using Fleetguard® COMPLEAT ES. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

Fill the cooling system with heavy-duty coolant and install the correct service filter. Use the following procedure in the appropriate Operation and Maintenance manual. Refer to Procedure 018-004 in Section V. (/qs3/pubsys2/xml/en/procedures/101/101-018-004.html)



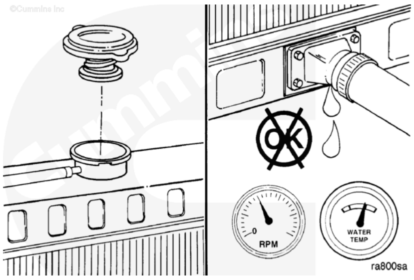
Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.

Note : Some radiators have two fill locations, both of which must be filled when the cooling system is drained.

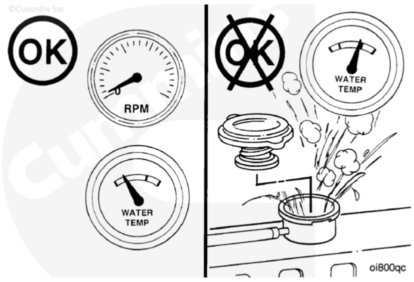


Install the radiator or expansion tank fill cap.

Operate the engine until it reaches a temperature of 82°C [180°F], and check for leaks.



Shut the engine OFF and allow to cool.



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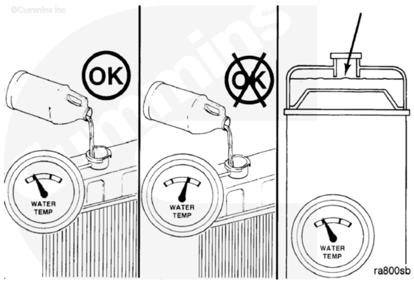

⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Remove the radiator or fill cap and check the coolant level. Refer to the OEM service manual instructions for recommendations. Add coolant if necessary. Do **not** overfill.

Last Modified: 08-Apr-2015